Tampere University
Tampere University of Applied Sciences

TIJO222 – Introduction to Data Science Code Clinic 1

[6 March 2026]

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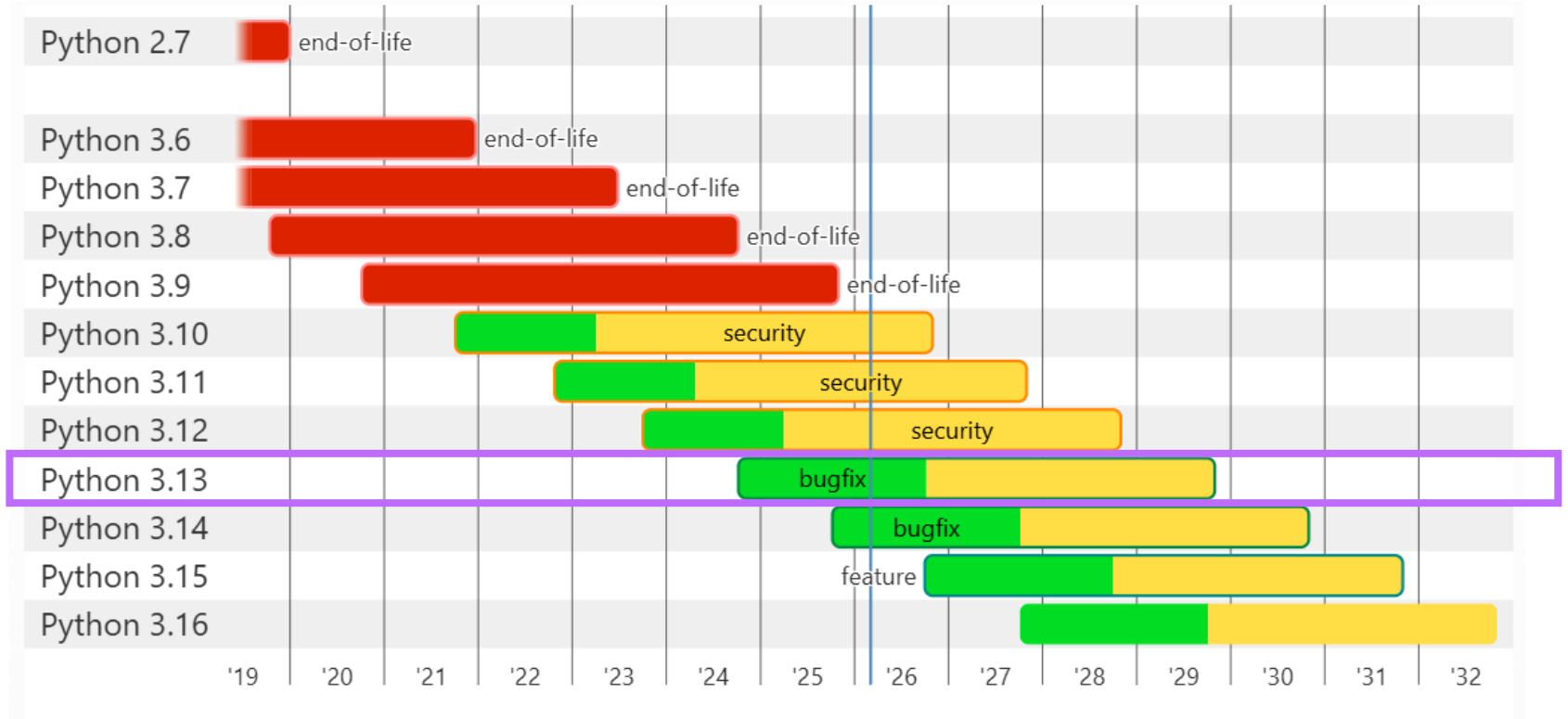
Setting Up Your Development Environment

- Python
- Selecting a development environment
- Why virtual environments are useful
- Selected environment demonstration

Make sure you have python installed in your laptop



https://www.python.org/downloads/

The Python version used by Plussa grader is **Python 3.13.5**.

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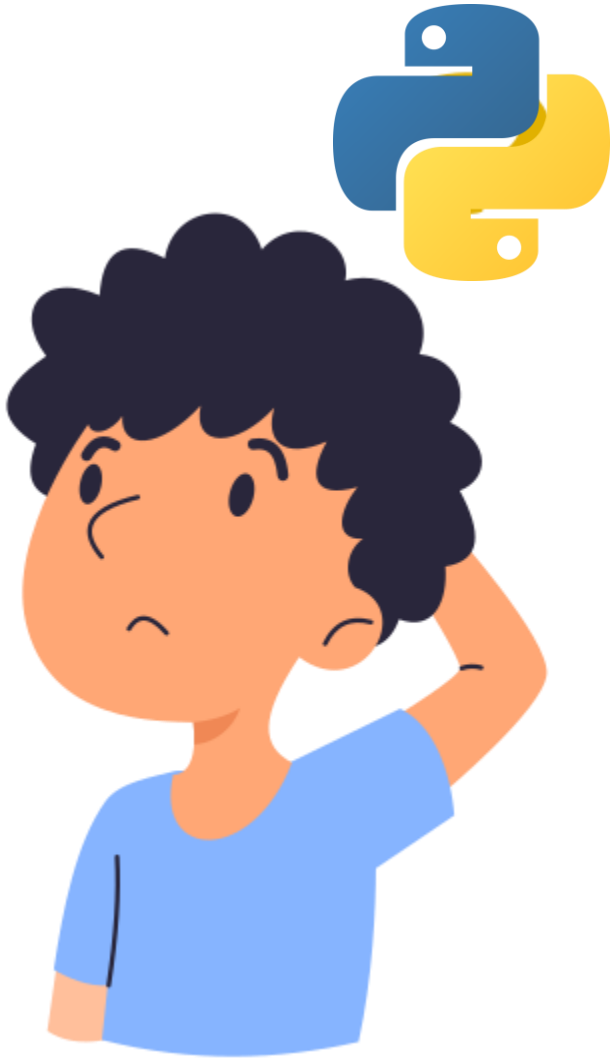


No python in your laptop?

That’s fine, you can download it.

Python installation guideline

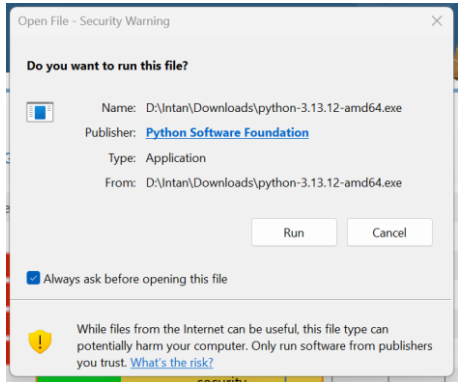
1. Go to https://www.python.org/downloads/
2. Choose your desired python version

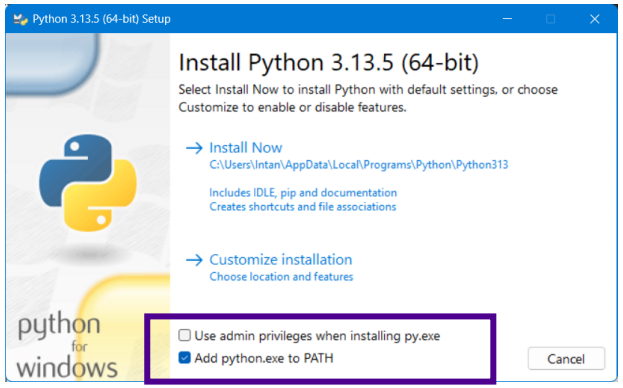
- Python 3.13.5 - June 11, 2025

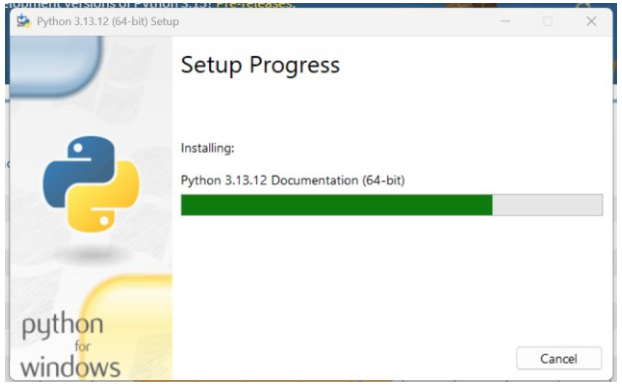
Note that Python 3.13.5 *cannot* be used on Windows 7 or earlier.

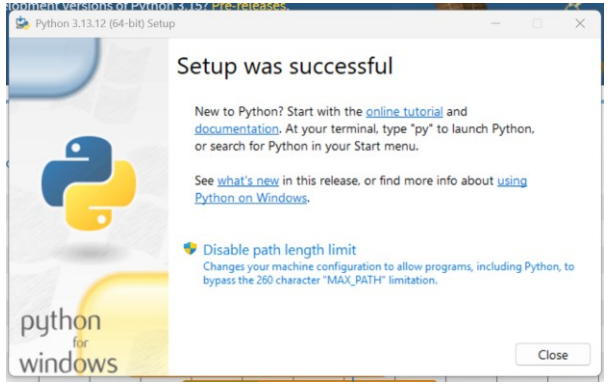
- Download Windows installer (64-bit)
- Download Windows installer (32-bit)
- Download Windows installer (ARM64)
- Download Windows embeddable package (64-bit)
- Download Windows embeddable package (32-bit)
- Download Windows embeddable package (ARM64)

3. Run the python-3.13.5-amd64.exe









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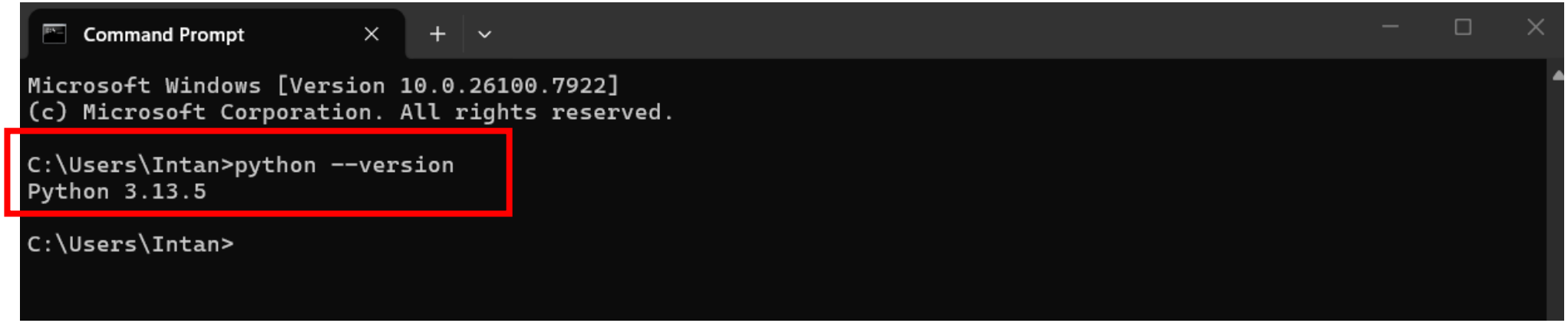
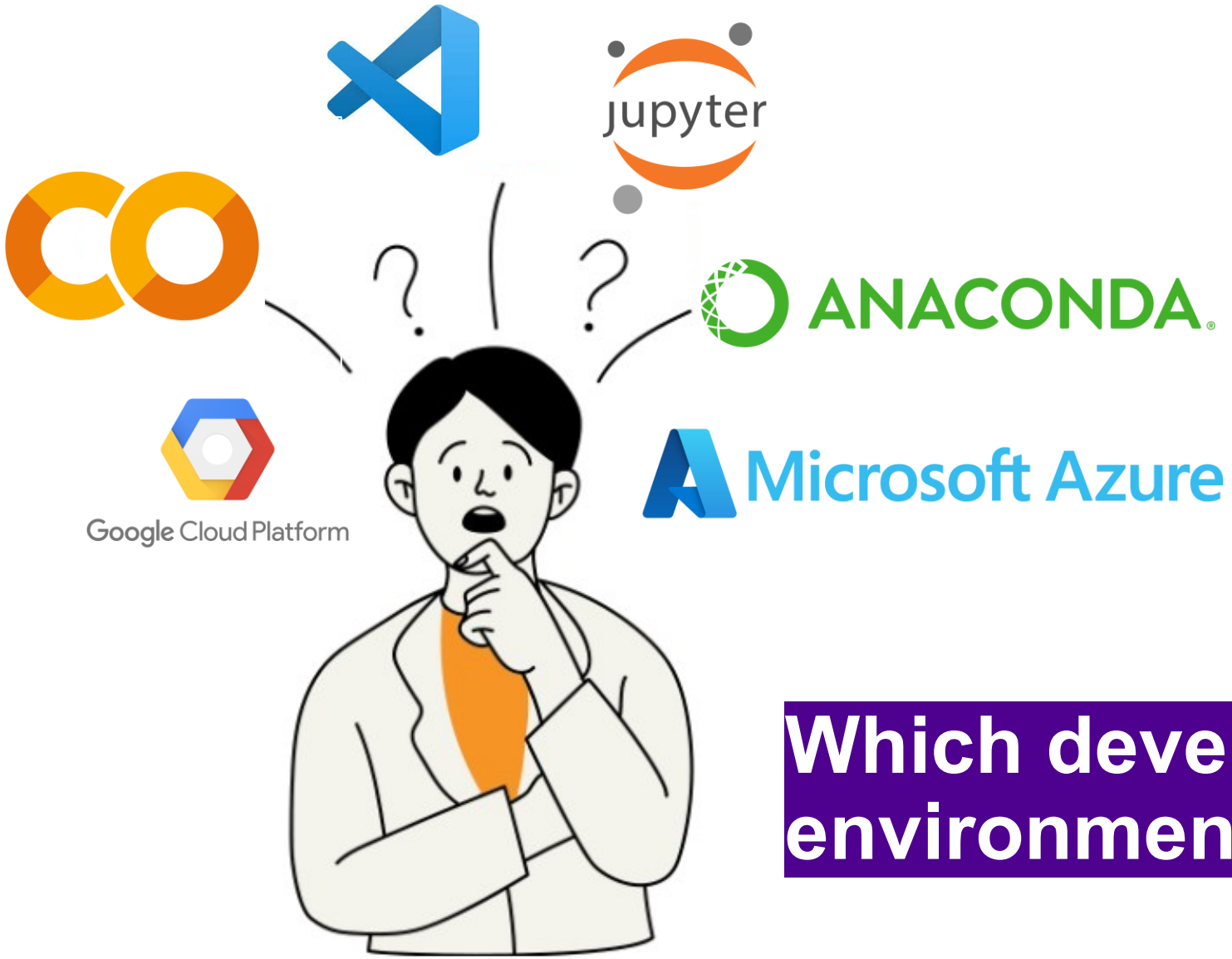
Check your installed python

1. Go to command prompt and write

```
python --version
```

2. You should be able to see the python version you just installed





Which development environment should I use?

Jupyter Notebook

An interactive web-based notebook for Python (combining code, text, and visualizations in one document) [.ipynb]

Pros
- Run code cell by cell, making experimentation easy
- Combines code, explanations, and visualizations in one place

Cons
- Running cells out of order can create confusing results
- Notebook files are harder to track/review with version control (Git)

Best for
- Data exploration and analysis
- Teaching and tutorials
- Documenting data science workflows

ANACONDA

A Python distribution designed for data science, including many libraries and tools.

Pros
- Includes Python, Jupyter, and many data science packages
- Easy environment management
- Beginner friendly

Cons
- Very large installation (Minimum 5 GB disk space to download and install)
- Sometimes installs packages you don't actually need

Best for
- Beginners in data science
- Users who want everything preinstalled

Visual Studio Code

A modern code editor with strong support for Python and Jupyter.

Pros
- Powerful development tools
- Low disk space required (< 200 MB)
- Supports both .py files and notebooks
- Many useful extensions
- Widely used in industry

Cons
- Requires installing extensions
- Slightly more setup for beginners

Best for
Students who want a professional coding environment

Google Colab

A cloud-based Jupyter notebook environment that runs in the browser.

Pros
- No installation required
- Easy to start immediately
- Free GPU access (limited)
- Easy sharing and collaboration

Cons
- Requires internet connection
- Session can reset (runtime disconnects)
- You need additional setup for submitting assignment

Best for
- Students who don't want to install software

Google Cloud Platform

Full cloud computing platforms from Google where you can build a custom data science environment.

Pros
- Very powerful and scalable
- Used in professional data engineering and ML workflows
- Can run large datasets and models

Cons
- Much more complex setup
- Requires cloud account and configuration
- Can cost money

Best for
advanced userslarge-scale data science projects

Microsoft Azure

Full cloud computing platforms from Microsoft where you can build a custom data science environment.

Pros
- Very powerful and scalable
- Used in professional data engineering and ML workflows
- Can run large datasets and models

Cons
- Much more complex setup
- Requires cloud account and configuration
- Can cost money

Best for
advanced userslarge-scale data science projects

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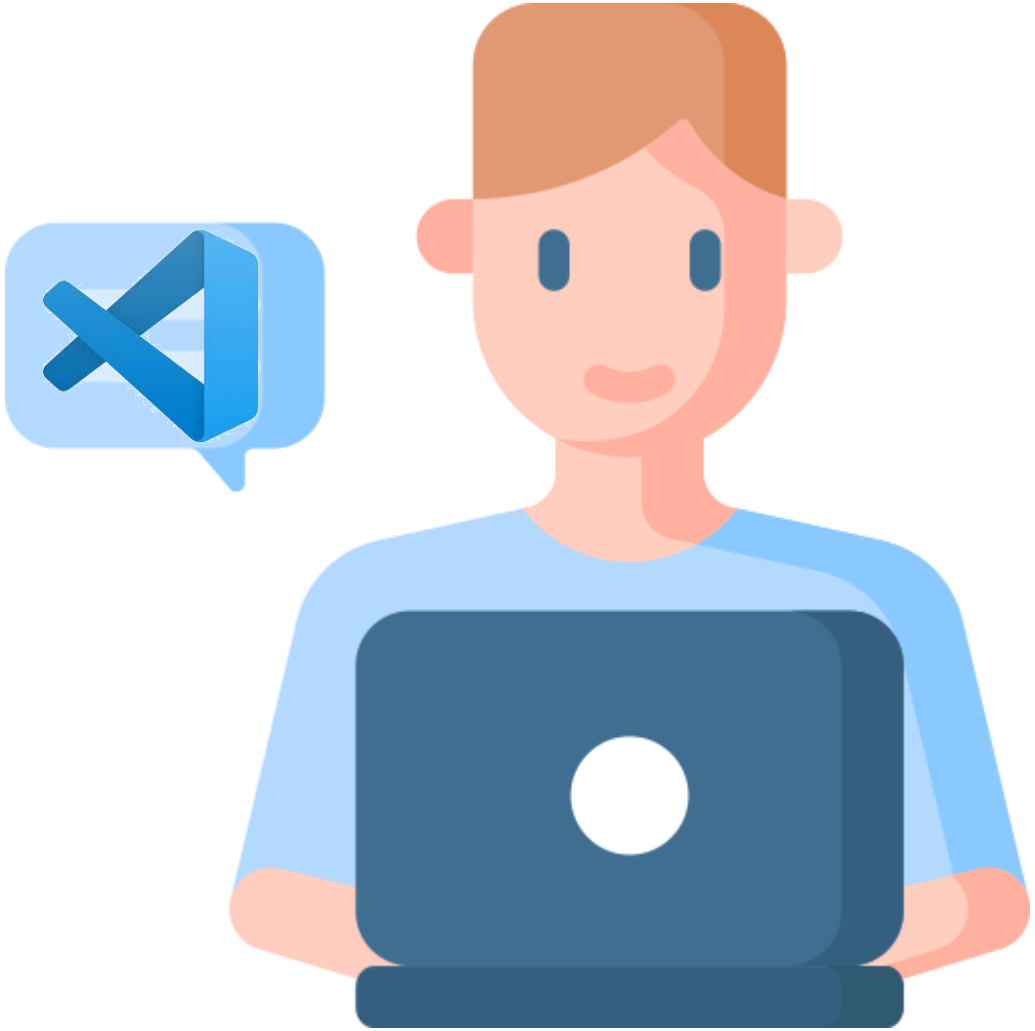
Install Visual Studio Code

- Go to : https://code.visualstudio.com/
- Click download for windows/mac
- Run the downloaded installer
- Search Visual Code in your laptop



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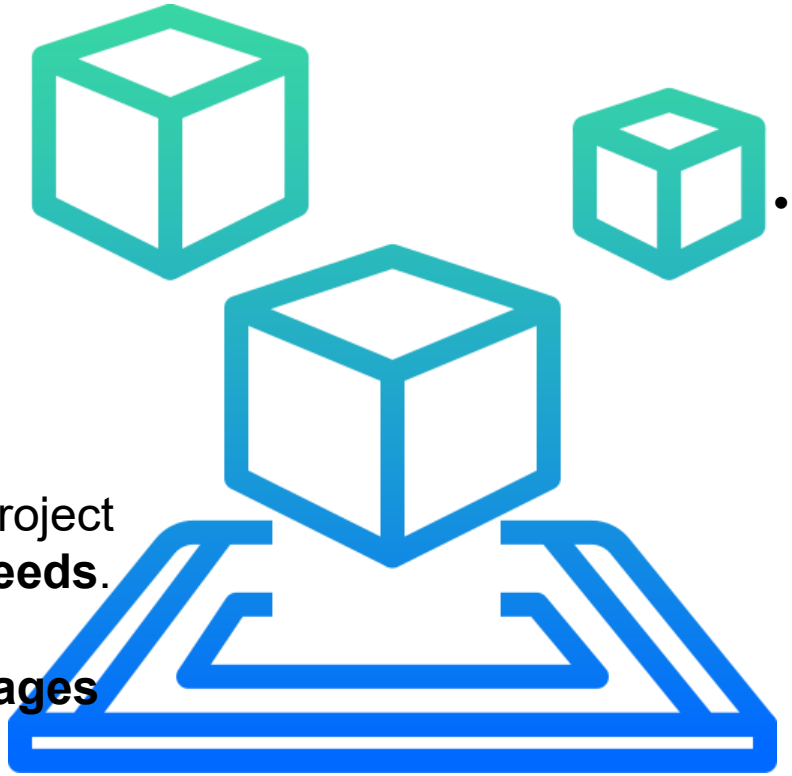
Why Creating Python Virtual Environment is Useful?

1. **Avoid Dependency Conflicts**
   - Different projects often require **different versions of the same package**.
   - If everything is installed in the same environment, packages may **overwrite each other** and break projects. A virtual environment keeps dependencies **separate per project**.

2. **Cleaner Project Management**
   - A virtual environment ensures that each project only contains **the packages it actually needs**.
   - Without isolation, your environment may contain **hundreds of unnecessary packages**
   - The project is easier to maintain.



3. **Reproducibility**
   - Research, development, and teaching often require **reproducible environments** (example: requirement.txt)
   - Other people can recreate the **exact same environment**, ensuring code runs identically.

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Setting Up Virtual Environment

- Open VS Code
- Install extension python and jupyter
- Open terminal and check your python

```
python --version
```

- Create virtual environment

```
python -m venv .\venv
```

- Activate your virtual environment

```
.\venv\Scripts\activate
```

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Install required package

- Install dependencies/packages

```
# install only one package
pip install notebook==7.5.3     #Semmantic Versioning
```

```
# install packages listed in a file
pip install -r ./requirements.txt
```

- Check all your installed package

```
pip list
```

Note: Check all available python version and latest version from pypi.org

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Let’s learn together!